

Surajit Tunga

Kolkata, West Bengal | +91-8972195682 | surajittunga2005@gmail.com
github.com/Surajit-Tunga | linkedin.com/in/surajittunga | surajit-five.vercel.app

Education

Guru Nanak Institute of Technology

B.Tech. in Electronics and Communication Engineering, CGPA: 8.5 (upto 5th semester)

Kolkata, West Bengal

2023 – Present

Technical Skills

Languages: Java, JavaScript, C, Python

Remote Sensing & GIS: Google Earth Engine, GIS, MIDAS, Sentinel-2, LISS-4

Web Technologies: React.js, Ejs, Node.js, Express.js

Database: MySQL, MongoDB, PostgreSQL

Core Concepts: Data Structures and Algorithms, Database Management Systems, Computer Networks, Object Oriented Programming

Tools: Git, GitHub, Postman, VS Code, MongoDB Compass, MySQL Workbench

Research Projects

LISM – Lunar Ice Survey and Mapping

Ongoing College Research Project

- Project Lead – processed Chandrayaan-2 DFSAR data using MIDAS and developed the complete web platform, covering both the data-processing and visualization pipeline.
- Developing a geospatial decision-support framework for detecting and characterizing potential subsurface ice in the lunar south polar region using Chandrayaan-2 DFSAR, OHRC, NASA ShadowCam, and LOLA DEM datasets.
- Working on multi-source geospatial data fusion, terrain analysis, landing-site assessment, and energy-aware rover traverse planning for future lunar exploration missions.
- Utilizing MIDAS, QGIS, GDAL, Rasterio, GeoJSON, and Python-based geospatial processing for planetary remote sensing workflows.
- Designing an interactive GIS dashboard for visualization of terrain, ice-confidence maps, landing sites, and rover traverses.

Dhrithi – Robust Change Detection, Monitoring, and Alert System on User-defined AOI using Multi-Temporal Sentinel-2 Satellite Imagery

Published Research | Journal of Geomatics | April 2026

- Project Lead – architected the end-to-end Web-GIS application using the MERN stack and Google Earth Engine workflows.
- Developed an AI-powered Web-GIS platform for automated land use/land cover (LULC) change detection using multi-temporal Sentinel-2 imagery and Google Earth Engine.
- Designed an automated pipeline for cloud and shadow masking, semantic segmentation, object-based change detection, and dynamic validation using Random Forest.
- Integrated geospatial technologies including Google Earth Engine, GeoJSON, Sentinel-2, LISS-4, GIS workflows, and MERN Stack for an end-to-end Web-GIS application.
- Published in the Journal of Geomatics (Vol. 20, No. 1, 2026).

Personal Projects

Resume Builder App | [Link](#)

- Developed a full-stack resume builder web application using the MERN stack with optimized database design for managing multiple user resumes and real-time editing features.

Achievements

IHMMC 2026 (International Level Hardware Model Making Competition) – *Winner (6th Position)*

March 2026

- Developed VitaSense, a non-invasive system that monitors vital health parameters like heart rate and oxygen levels in real time.
- Team Lead & Developed the complete web platform.

Smart Bengal Hackathon – *Finalist*

May 2025

- Developed a prototype for detection of person talking on mobile in no-mobile zone, collaborating with a 4-member team.
- Team Lead & Led backend development.

Certifications

Programming in Java, NPTEL – IIT Kharagpur – *Elite + Gold Certification (Top 1% Learner)*

October 2025

Introduction to Database Systems, NPTEL – IIT Madras – *Elite Certification*

May 2025

Area of Interest

Remote Sensing and GIS